

Jonathan Coulter

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Education

Master of Science in Computer Science

University of Pittsburgh

Jan 2025 - Dec 2026

4.0 / 4.0

- Relevant Coursework: Algorithm Design, Natural Language Processing, Human-Computer Interaction, Interface Design

Bachelor of Science in Computer Science

University of Pittsburgh – Minors in Mathematics & Economics

Aug 2022 - Dec 2025

3.97 / 4.0

- Relevant Coursework: Data Structures & Algorithms, Web Development, Software Quality Assurance, Operating Systems

Technical Skills

Languages: Java, Python, C, C++, JavaScript, TypeScript, HTML/CSS, Haskell, Rust, Swift

Frameworks: Flask, React, Spring Boot, Angular, Node.js, jQuery, JUnit

Tools: Git, Docker, Firebase, Google Cloud Platform, Linux, Excel, Jupyter

Experience

Software Engineer Intern

Bank of New York

Jun 2025 – Aug 2025

Pittsburgh, PA

- Built a real-time monitoring system aggregating 2M+ records across legacy and modern payment platforms for faster diagnostics
- Achieved 90% reduction in query latency by optimizing the Spring Boot backend API using dynamic SQL query building
- Developed an AI agent using BNY's Eliza platform to generate, execute, and summarize SQL queries into actionable responses, achieving 70% faster data retrieval for non-technical staff

Software Engineer Intern

Carnegie Mellon University

May 2024 – Dec 2024

Pittsburgh, PA

- Developed the UI of Sway, a chatroom that depolarizes conversations through AI moderation, using Bootstrap and jQuery to transform it into a modern, responsive platform
- Overhauled server logic to integrate a real-time Firebase database, allowing for modular configuration upon deployment
- Engineered an administrative dashboard using WebSockets to allow for live monitoring of platform metrics during studies

Data Science Research Fellow

Center for Analytical Approaches to Social Innovation

Aug 2023 – Apr 2024

Pittsburgh, PA

- Leveraged Python and pandas to structure data involving the fragmented Allegheny County police-citizen complaint process
- Integrated data from 130+ municipalities into an interactive GIS storymap, which now empowers the community by making the latest police accountability data more accessible and actionable for residents of Allegheny County

Projects

whentlymeet.com | Flask, React, Google Calendar API, PostgreSQL, Docker, NGINX

- Launched a full-stack group scheduling web app that attracted 100+ users, enabling creation and sharing of availability polls
- Integrated Google Calendar API to autofill availability, powering 300+ availability responses with minimal user input
- Designed and optimized a mobile-first React UI, driving 50+ events created and seamless scheduling across devices

Echo | Flask, Spotify API, OpenCV

- Developed a full-stack web app integrated with the Spotify API to generate 20+ song playlists for the user based on their mood
- Detected and pre-processed live user facial images in OpenCV and utilized a DeepFace model for mood analysis
- Built a RESTful API with Flask to handle user authentication, data processing, and interaction with APIs

Leadership

Undergraduate Teaching Assistant & Peer Tutor

University of Pittsburgh

Jan 2024 – Present

Pittsburgh, PA

- Led 25 students through weekly recitation discussions and exercises to clarify Data Structures & Algorithms concepts
- Hosted office hours available to 150+ students where I found unique explanations that cater to diverse learning styles, enhancing their grasp of programming assignments, practice exercises, and problem-solving techniques

CSC Board Member

Pitt Computer Science Club

Apr 2025 – Present

Pittsburgh, PA

- Co-led Bit/Byte, a mentorship program connecting 120+ new students with mentors leading to 50% receiving internship offers

Section Leader

Stanford University Code in Place

Apr 2025 – May 2025

Stanford, CA - Remote

- Led a weekly virtual section of 15 students through an international summer session of Stanford's CS 106A on Python programming